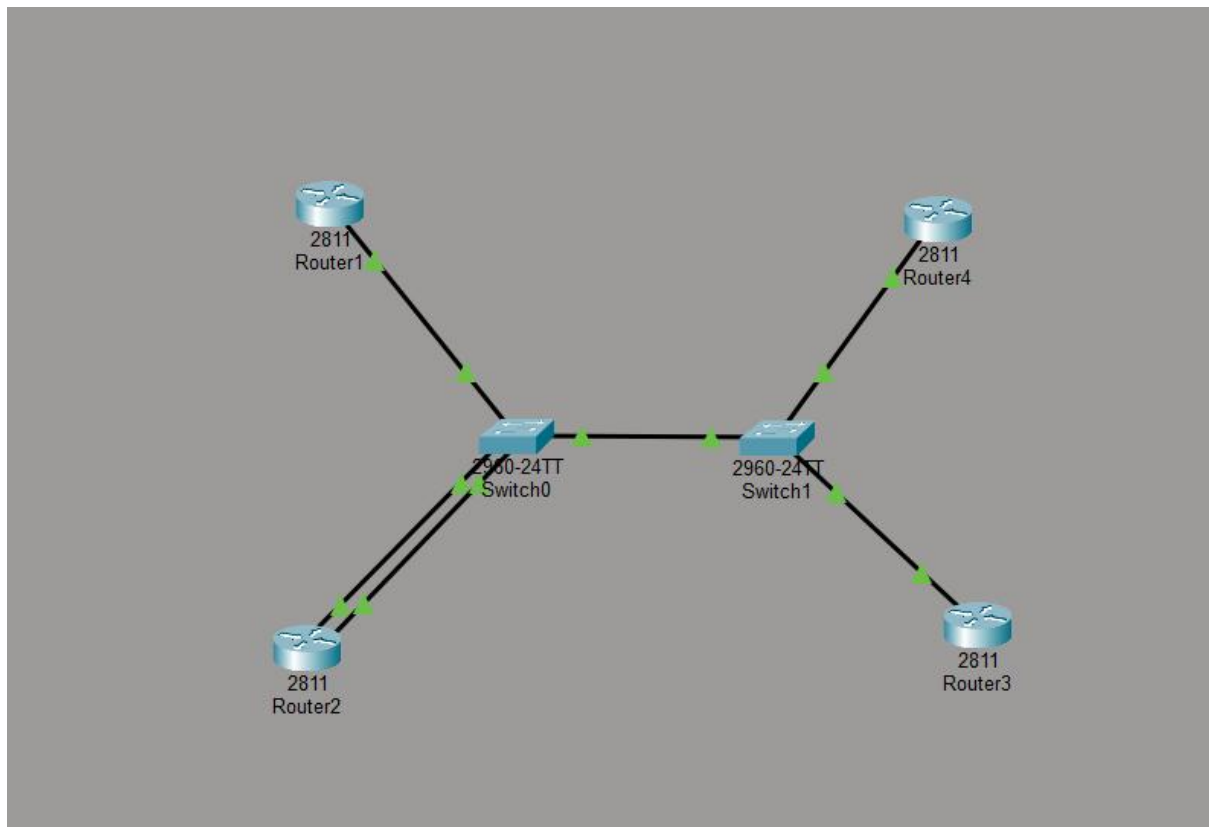


# JJ H Packet tracer week 2 20260176

## Contents

|                                                                      |   |
|----------------------------------------------------------------------|---|
| JJ H Packet tracer week 2 20260176.....                              | 1 |
| Topology of this week 2 prac .....                                   | 1 |
| Step 5: Verify switch mac address table .....                        | 2 |
| Step 6: Note down mac address of each interface on each router ..... | 3 |
| Step 7: Verify connectivity between routers using ping.....          | 4 |
| Step 8: Dynamically learned MAC addresses on the switch .....        | 5 |

## Topology of this week 2 prac



## Ip addresses

Switch 0 192.168.1.11 255.255.255.0

Switch 1 192.168.1.12 255.255.255.0

Router 1 192.168.1.1 255.255.255.0

Router 2 192.168.1.2 255.255.255.0  
Router 2 Port F0/1 192.168.2.1 255.255.255.0  
Router 3 192.168.1.3 255.255.255.0  
Router 4 192.168.1.4 255.255.255.0

## Step 5: Verify switch mac address table

Switch 1

```
Switch0

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       000c.854e.3c01    DYNAMIC   Fa0/2
1       0060.4702.8201    DYNAMIC   Fa0/24
1       0090.2198.0001    DYNAMIC   Fa0/1
1       0090.2198.0002    DYNAMIC   Fa0/3
1       00d0.ba83.9e18    DYNAMIC   Fa0/24
1       00d0.ffa0.b102    DYNAMIC   Fa0/24
1       00e0.a339.70c6    DYNAMIC   Fa0/24
Switch#
```

## Switch 2

```

Switch1
Physical Config CLI Attributes
Switch(config-if)#no shutdown
Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, chan
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       000c.854e.3c01    DYNAMIC     Fa0/24
1       000c.8558.cedb    DYNAMIC     Fa0/24
1       0060.4702.8201    DYNAMIC     Fa0/4
1       0090.2198.0001    DYNAMIC     Fa0/24
1       00d0.bcd0.6a18    DYNAMIC     Fa0/24
1       00d0.ffa0.b102    DYNAMIC     Fa0/3
Switch#

```

## Step 6: Note down mac address of each interface on each router

### Router 1

| Device Name: Router1                                                                              |      |      |                |              |                |
|---------------------------------------------------------------------------------------------------|------|------|----------------|--------------|----------------|
| Custom Device Model: 2811 IOS15                                                                   |      |      |                |              |                |
| Hostname: Router                                                                                  |      |      |                |              |                |
| Port                                                                                              | Link | VLAN | IP Address     | IPv6 Address | MAC Address    |
| FastEthernet0/0                                                                                   | Up   | --   | 192.168.1.1/24 | <not set>    | 000C.854E.3C01 |
| FastEthernet0/1                                                                                   | Down | --   | <not set>      | <not set>    | 000C.854E.3C02 |
| Vlan1                                                                                             | Down | 1    | <not set>      | <not set>    | 00D0.BC31.28C5 |
| Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1 |      |      |                |              |                |

### Router 2

Device Name: Router2  
Custom Device Model: 2811 IOS15  
Hostname: Router

| Port            | Link | VLAN | IP Address | IPv6 Address | MAC Address    |
|-----------------|------|------|------------|--------------|----------------|
| FastEthernet0/0 | Up   | --   | <not set>  | <not set>    | 0090.2198.0001 |
| FastEthernet0/1 | Up   | --   | <not set>  | <not set>    | 0090.2198.0002 |
| Vlan1           | Down | 1    | <not set>  | <not set>    | 00E0.F99E.2492 |

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router2

### Router 3

Device Name: Router3  
Custom Device Model: 2811 IOS15  
Hostname: Router

| Port            | Link | VLAN | IP Address     | IPv6 Address | MAC Address    |
|-----------------|------|------|----------------|--------------|----------------|
| FastEthernet0/0 | Down | --   | <not set>      | <not set>    | 00D0.FFA0.B101 |
| FastEthernet0/1 | Up   | --   | 192.168.1.3/24 | <not set>    | 00D0.FFA0.B102 |
| Vlan1           | Down | 1    | <not set>      | <not set>    | 000C.CFB4.7A33 |

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router3

### Router 4

Device Name: Router4  
Custom Device Model: 2811 IOS15  
Hostname: Router

| Port            | Link | VLAN | IP Address     | IPv6 Address | MAC Address    |
|-----------------|------|------|----------------|--------------|----------------|
| FastEthernet0/0 | Up   | --   | 192.168.1.4/24 | <not set>    | 0060.4702.8201 |
| FastEthernet0/1 | Down | --   | <not set>      | <not set>    | 0060.4702.8202 |
| Vlan1           | Down | 1    | <not set>      | <not set>    | 0090.0C9E.4622 |

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router4

## Step 7: Verify connectivity between routers using ping

Router 1 ping test to all other routers, failed at the second connection of router 2, first one successful

```
Router#ping 192.168.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#ping 192.168.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping 192.168.1.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.3, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router#ping 192.168.1.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.4, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#
```

router 1 pinging both switches successfully

```
Router#ping 192.168.1.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.11, timeout is 2 seconds:
..!!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.1.12

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.12, timeout is 2 seconds:
..!!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/2 ms

Router#
```

## Step 8: Dynamically learned MAC addresses on the switch

```
Switch0
Switch>enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.854e.3c01    DYNAMIC Fa0/2
1       0090.2198.0001    DYNAMIC Fa0/1
1       0090.2198.0002    DYNAMIC Fa0/3
1       00d0.ba83.9e18    DYNAMIC Fa0/24
1       00e0.a339.70c6    DYNAMIC Fa0/24
Switch#
```

### Step 9: Clear table, check again

```
Switch#clear mac address-table
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       00d0.ba83.9e18    DYNAMIC Fa0/24
```

Now only shows 1, but adds the others over a couple seconds

### Step 10, show ip route

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0

Router#

```

---

**Step 11** wants me to configure IP addresses which have already been done and all connect fine. Though I understand that Step 12 wants to see that they all show up after I use "Show ip route" Which they do.

## Step 13 and 14:

Tried to configure static routes, however I believe they only work on serial connections

```

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.1.11
% Incomplete command.
Router(config)#ip route 192.168.1.11 255.255.255.0
% Incomplete command.
Router(config)#ip route 192.168.1.11 255.255.255.0 10.1.1.2
%Inconsistent address and mask
Router(config)#

```

---

I did try to make a static route from the router 1 to switch 0

## Reflection question 15

After clearing the MAC address table I found out that the MAC addresses reappear over a few seconds, I believe this is because the addresses are automatically found via being connected in the network.